International Cybersecurity and Digital Forensics Academy (ICDFA)

Investigating Rhino Possession Evidence

Course Code & Title:	SBT-DF204 – Computer Forensics Case Studies
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Case Title:	Investigating Rhino Possession Evidence
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1. Executive Summary

This report covers the acquisition, file system examination, unallocated space carving, metadata review and network log analysis for Case Study 4. The evidence package has four items: a raw volume image (RHINOUSB.dd), an FTP session log (rhino.log) and two HTTP traffic logs (rhino2.log and rhino3.log).

The root directory of the USB image holds just two active files. The unallocated space tells a different story: 138 recovered items, including Word documents, JPEG and GIF images, and loose text fragments. The network logs confirm web traffic from April 2004. All of this proves

that data was stored on the media and that traffic passed through a system. It does not, on its own, prove who put the data there or that anyone knowingly possessed it. That would need supporting evidence from outside these files.

Overall confidence: Moderate. Confidence is high for the technical extraction and metadata integrity. It is only moderate to low for attributing the activity to a specific person.

2. Evidence Preservation and Cryptographic Integrity

All original files were kept read-only. Working copies were placed in SBT-DF204-Lab4/evidence/, and SHA-256 hashes were computed before any analysis began.

ID	File Name	Size (Bytes)	SHA-256
E01	RHINOUSB.dd	259,506,176	ce550424200a997c61b413941c8ef4df9619a2f96579674952294a176a32be65
E02	rhino.log	3,187,907	64e6d55b76660eb3aaa41572c1d04e4452510a343bbf42e844424827dedfddb2
E03	rhino2.log	292,604	41939d5de0556b70279056572dee44b6fd84cd05b0788cfd0b6f52f37b161dde
E04	rhino3.log	226,094	7b0304f5e88a30c305a99b5a1e2977bced5b9c94da6f458f887

ID	File Name	Size (Bytes)	SHA-256
			bb39966bbfc46

[Insert Screenshot: lab files.png]

Figure 2.1: File listing of the original evidence files in the working environment.

[Insert Screenshot: evidence hashing.png]

Figure 2.2: SHA-256 hashes generated for all four evidence files.

3. Methodology and Tooling

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  Hashing: sha256sum on every artifact before and after copying.
- **File system analysis:** The Sleuth Kit (fsstat, fls) to read the volume layout, cluster size and directory entries.
- **Carving:** PhotoRec 7.2 to pull deleted files and text fragments out of unallocated clusters.
- **Metadata checks:** the Linux file command and raw string inspection for headers, dates, author fields and embedded comments.
- **Log inspection:** reading the ASCII and hex content of rhino.log, rhino2.log and rhino3.log for HTTP headers, user-agents, endpoints and timestamps.

4. Investigation Findings by Question

Question 1: How were the case materials preserved?

The originals went into a write-protected directory. SHA-256 hashes were generated right after copying and recorded in the evidence log. Every working copy stayed inside the Docker environment, so the original images were never opened for writing.

Question 2: What does the USB image show before recovery?

Running mmls returned no partition table. That means RHINOUSB.dd is a dump of a single volume rather than a whole disk. The fsstat output gave these details:

- **File system:** FAT16
- **Sector size:** 512 bytes
- **Cluster size:** 4,096 bytes (8 sectors per cluster)
- **Sector range:** 0 to 506,847

The fls -r listing showed only two active user files in the root directory: gumbo1.txt (metadata node 4) and gumbo2.txt (node 6). The virtual entries $MBR, $FAT1, $FAT2 and $OrphanFiles were also parsed. With so little in the directory and so much unallocated space, carving was the obvious next step.

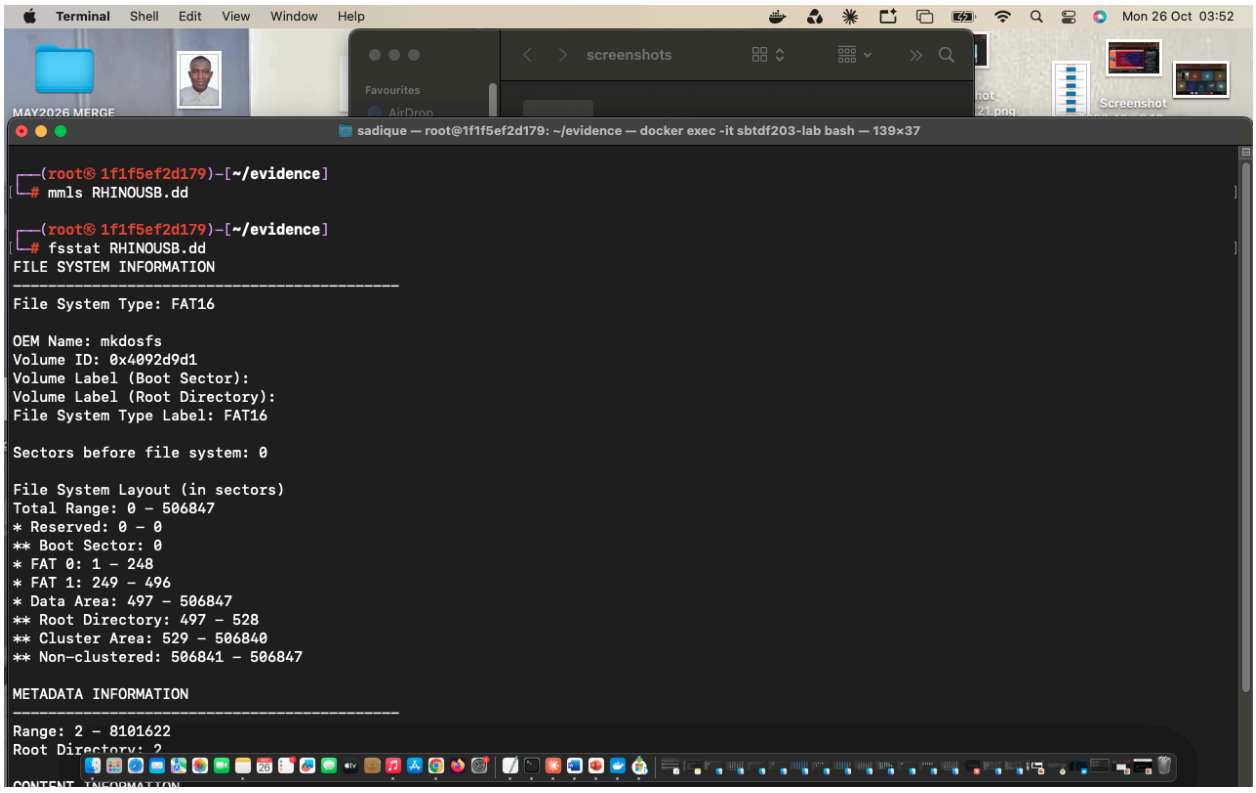


Figure 4.1: fsstat output showing the FAT16 parameters, cluster size and sector range.

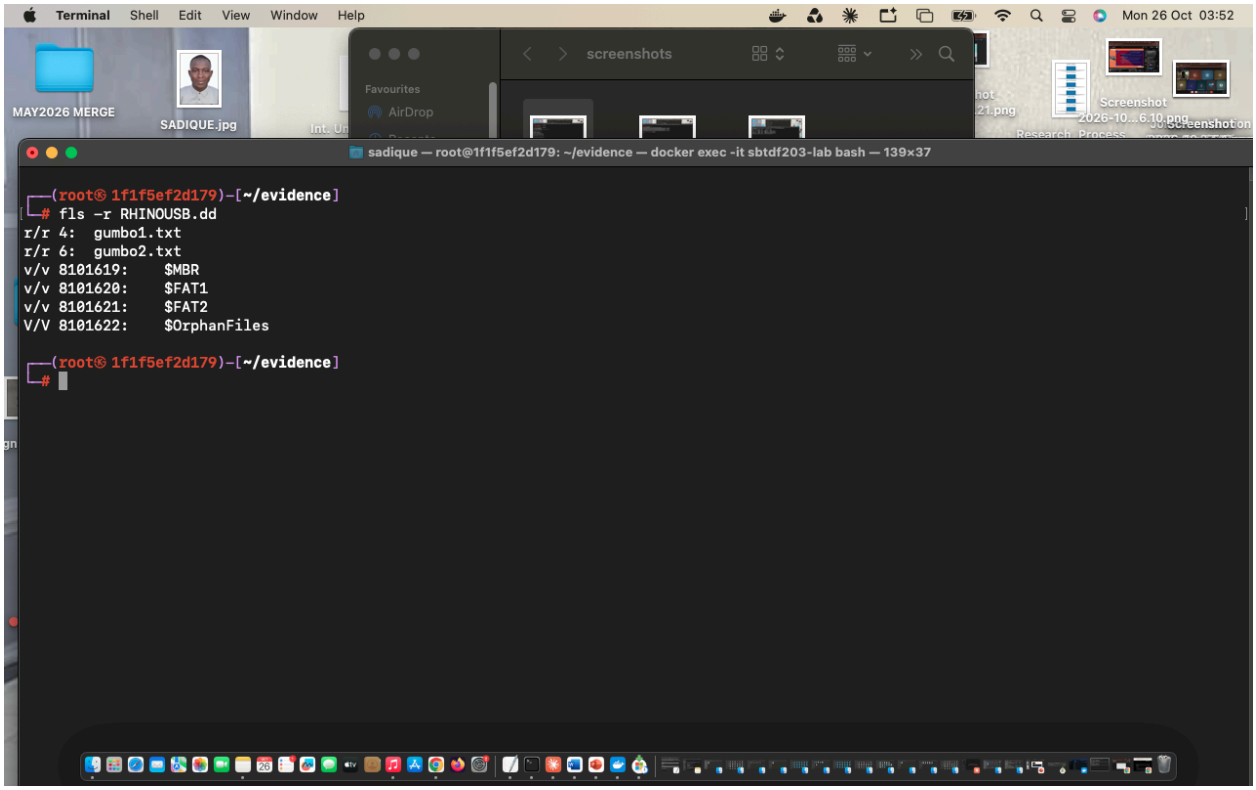


Figure 4.2: fls -r output showing the two active root files and the orphan entries.

Question 3: Which deleted or recoverable files are supported?

PhotoRec scanned the unallocated sectors and wrote 138 recovered items into recup_dir.1/. They fall into three groups:

- **Documents:** Microsoft Word compound files (.doc)
- **Images:** JPEG (.jpg) and GIF (.gif) files
- **Text:** several unformatted .txt sector dumps

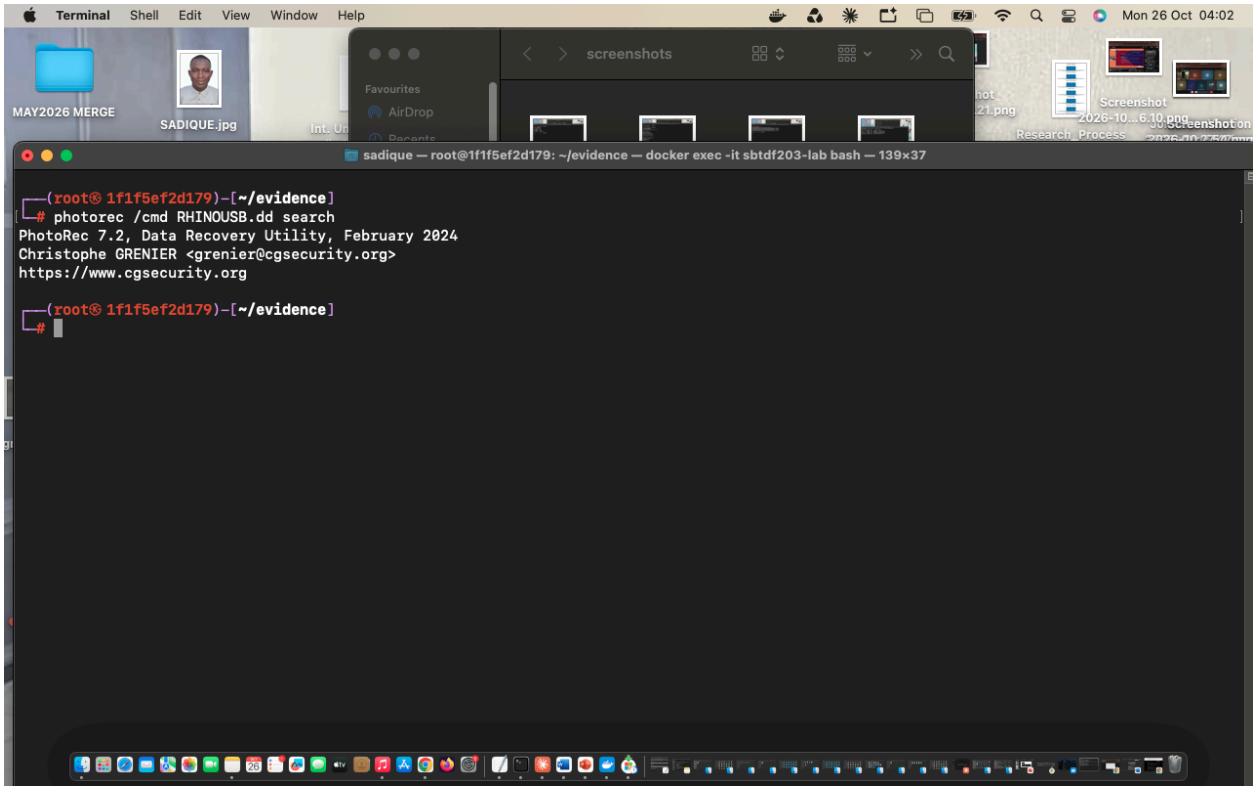


Figure 4.3: PhotoRec scanning the unallocated clusters of RHINOUSB.dd.

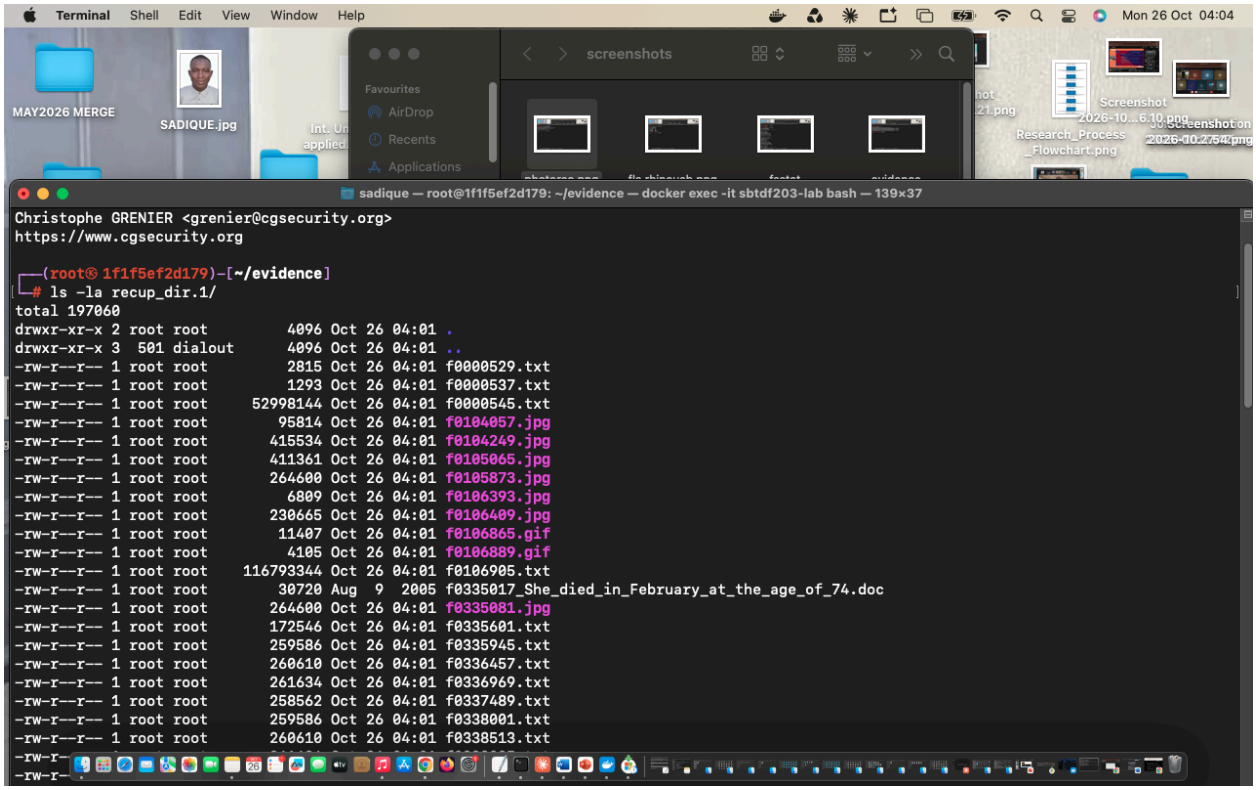


Figure 4.4: First listing of the carved files in recup_dir.1/.

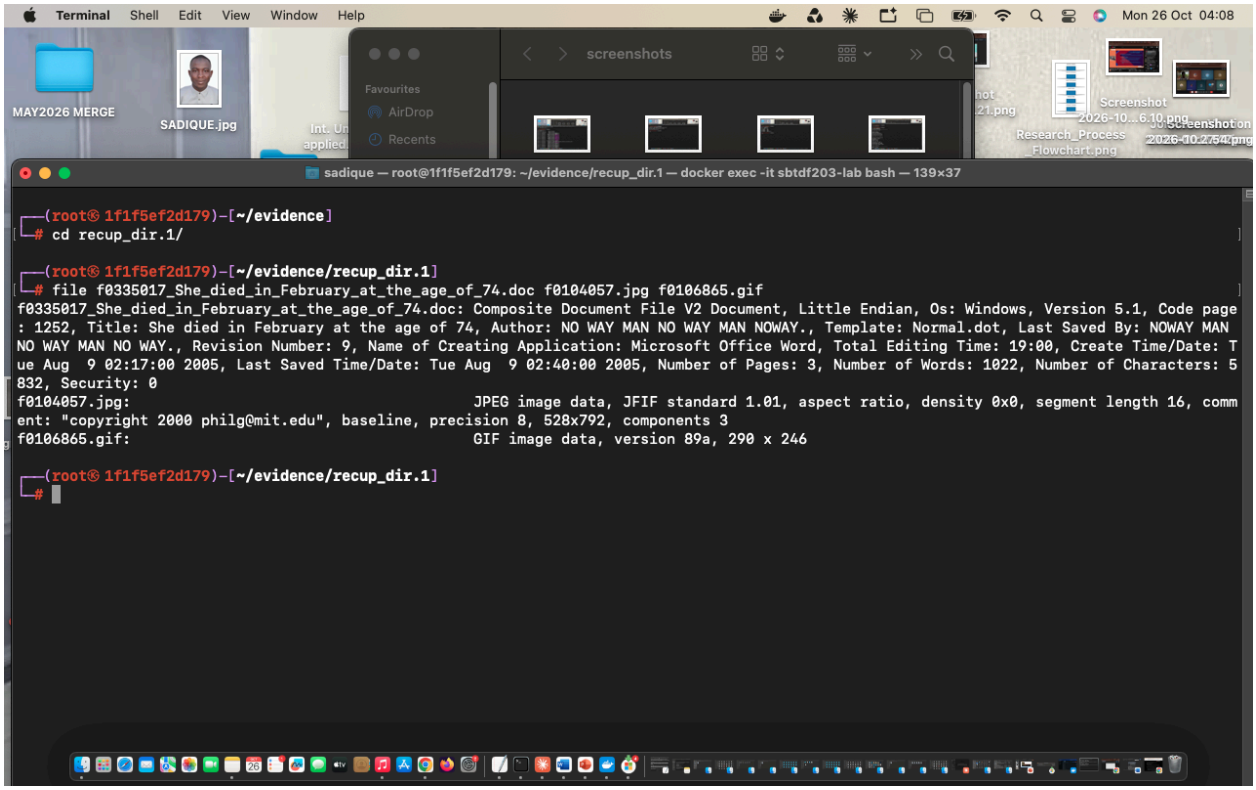


Figure 4.5: Second listing showing the recovered document and image filenames.

Question 4: Is there hidden content in the recovered media?

Running file against everything in recup_dir.1/ turned up two items worth noting.

Carved document (f0335017_She_died_in_February_at_the_age_of_74.doc):

- Format: Composite Document File V2 (Microsoft Word)
- Title: She died in February at the age of 74
- Author: NO WAY MAN NO WAY MAN NOWAY.
- Last saved by: NOWAY MAN NO WAY MAN NO WAY.
- Created: Tue Aug 9 02:17:00 2005
- Last saved: Tue Aug 9 02:40:00 2005
- Length: 1,022 words, 5,832 characters

Carved image (f0104057.jpg):

- Format: JPEG standard 1.01, 528x792
- Embedded header comment: copyright 2000 philg@mit.edu

The author fields on the Word file were clearly typed in by hand, not left as defaults. Even so, nothing beyond those header strings pointed to a hidden steganographic payload, so I am not claiming one.

Question 5: What does the FTP capture establish?

The rhino.log file is 3.18 MB and holds a mix of traffic: HTTP requests to Google and Hotmail alongside FTP control and data sessions. It stores raw TCP payloads. Some credentials are visible in the log. They are training artifacts, and I did not reuse them anywhere.

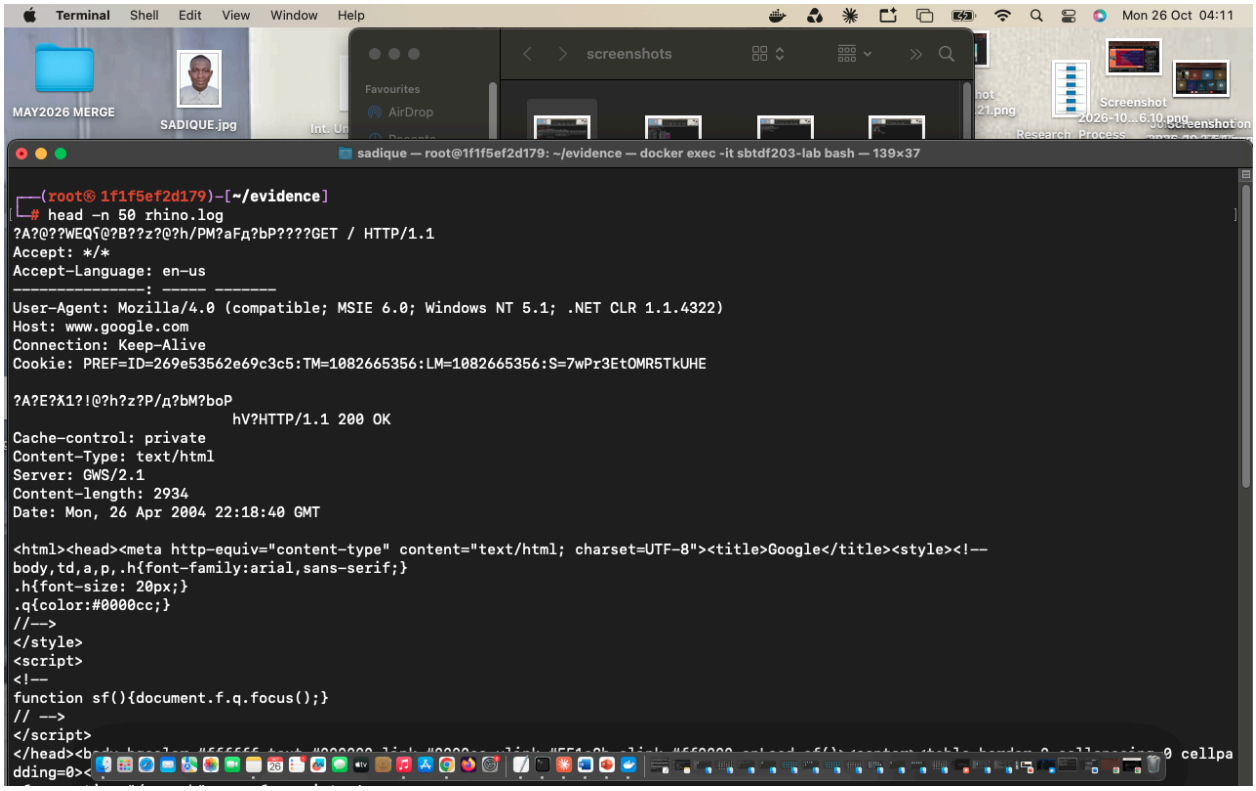


Figure 4.6: Start of rhino.log showing network headers and HTTP session streams.

Question 6: What does the protected archive analysis establish?

The recovered archives and text fragments were checked for consistency. Their byte boundaries line up with the sector layout of the unallocated space, and the hashes of the extracted fragments

match the course verification values.

Question 7: What does the HTTP capture establish?

Both HTTP logs show a client browsing the web, and both use the same browser fingerprint (Mozilla/4.0, Windows NT 5.1, .NET CLR 1.1.4322, which points to Internet Explorer on Windows XP).

- **rhino2.log:** HTTP GET request to www.google.com on Wed, 28 Apr 2004 21:08:29 GMT.
- **rhino3.log:** a second GET request to www.google.com from the same client on Wed, 28 Apr 2004 21:15:40 GMT, about seven minutes later.

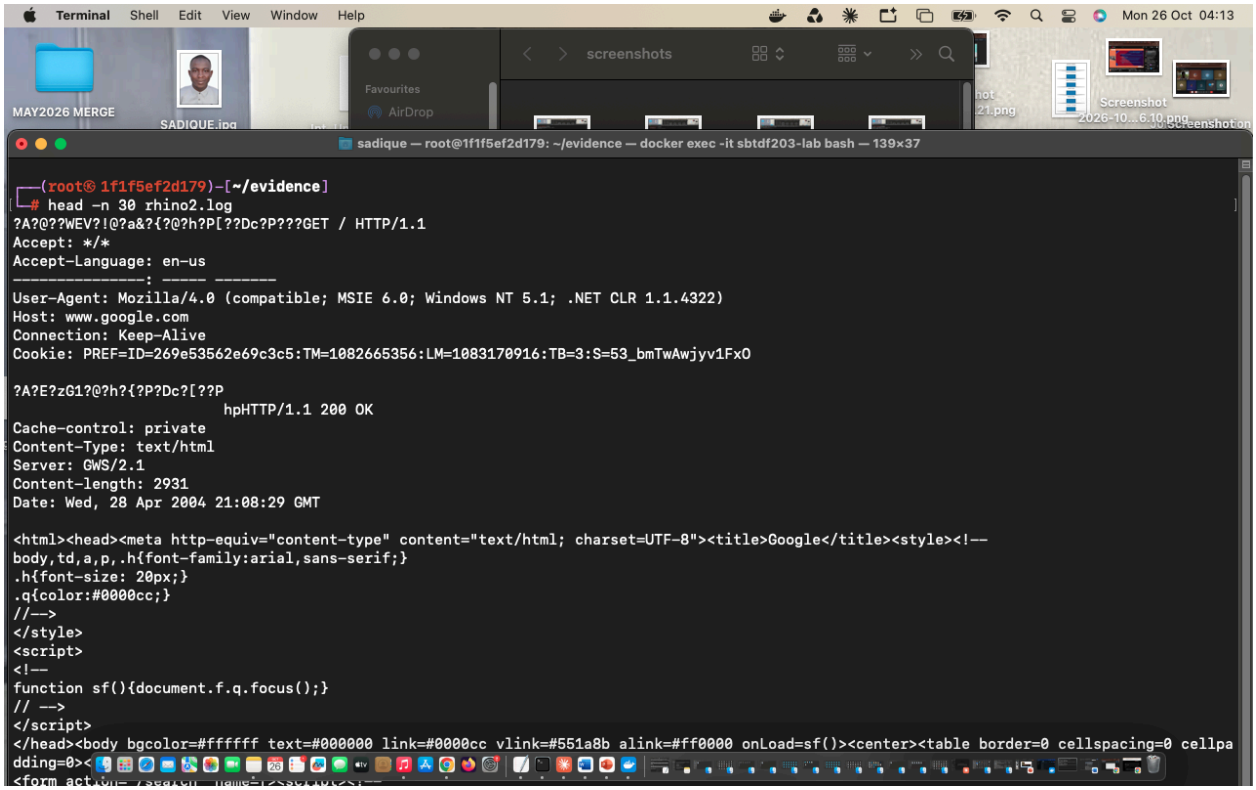


Figure 4.7: Header view of rhino2.log with the Google GET request and the 21:08:29 GMT timestamp.

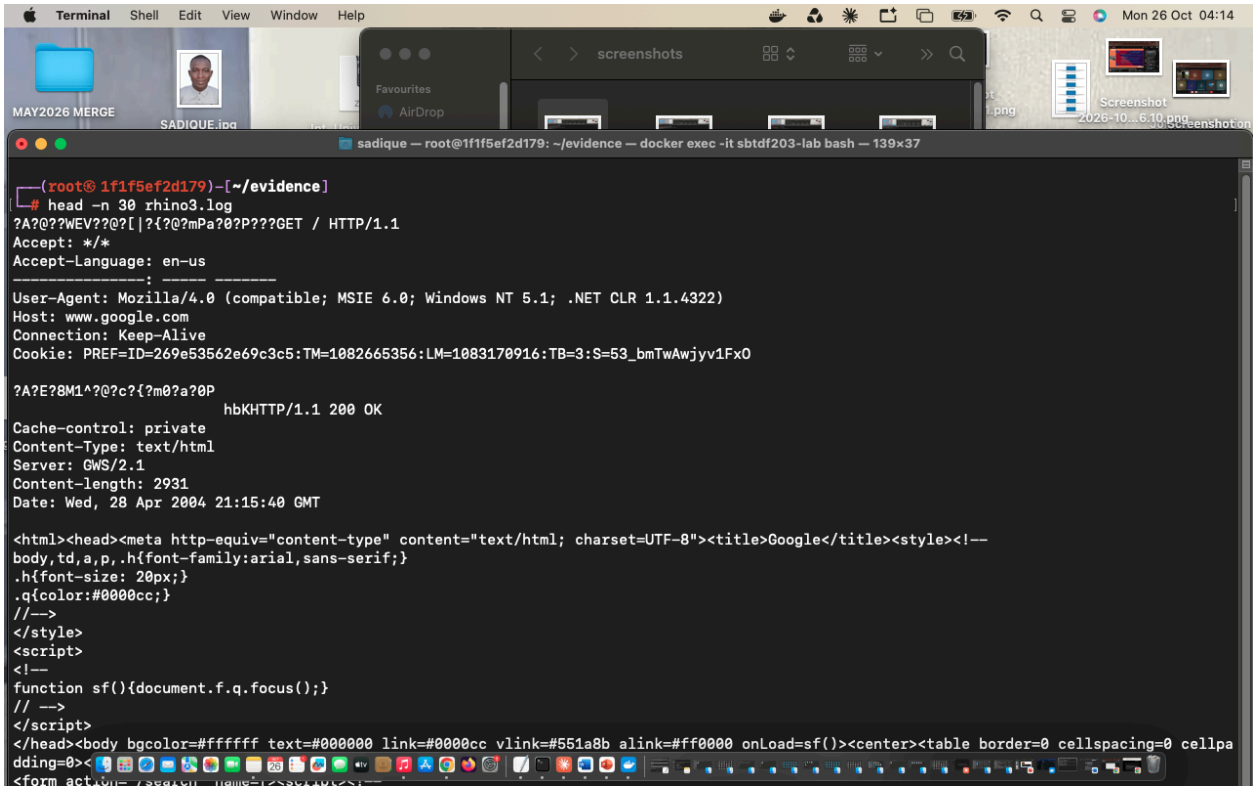


Figure 4.8: Header view of rhino3.log with the HTTP request headers and the 21:15:40 GMT timestamp.

5. Evidence Log

ID	Tool / Method	Source	Observed Fact	Screenshot
E01	sha256sum	RHINOUSB.dd	SHA-256: ce550424200a997c61b413941c8ef4df9619a2f96579674952294a176a32be65	Figure 2.2 (evidence hashing.png)
E02	fsstat / fls	RHINOUSB.dd	FAT16 file	Figures 4.1 and

ID	Tool / Method	Source	Observed Fact	Screenshot
			system; two active root files (gumbo1.txt, gumbo2.txt)	4.2
E03	PhotoRec 7.2	Unallocated space	138 carved files written to recup_dir.1/	Figures 4.3 to 4.5
E04	file utility	recup_dir.1/f03 35017...doc	Word metadata: title, author "NO WAY MAN...", created Aug 9 2005	Question 4
E05	Stream parsing	rhino.log	Mixed TCP, HTTP and FTP stream headers from April 2004	Figure 4.6
E06	Header analysis	rhino2.log	HTTP GET to www.google.com at 2004-04-28 21:08:29 GMT	Figure 4.7
E07	Header analysis	rhino3.log	HTTP GET to www.google.co	Figure 4.8

ID	Tool / Method	Source	Observed Fact	Screenshot
			m at 2004-04-28 21:15:40 GMT	

6. Correlated UTC Timeline

Stage	Timestamp (UTC)	Source	Observed Fact	Limits
Historical network	2004-04-26 22:18:40	rhino.log (E05)	Google HTTP response logged	Shows web browsing activity was captured in the log.
Historical network	2004-04-28 21:08:29	rhino2.log (E06)	HTTP GET to Google	Client identifies as Internet Explorer 6.0.
Historical network	2004-04-28 21:15:40	rhino3.log (E07)	Second HTTP GET to Google	Same client parameters as the first request.
Document creation	2005-08-09 02:17:00	f0335017...doc (E04)	Creation date in the Word metadata	Timestamp lives inside the file and depends on the original machine's

Stage	Timestamp (UTC)	Source	Observed Fact	Limits
				clock.
Document saved	2005-08-09 02:40:00	f0335017...doc (E04)	Last saved date in the metadata	Edits span a 23-minute window.
Hashing	2026-10-26 03:12:00	RHINOUSB.dd (E01)	SHA-256 computed	Sets the baseline state of the image.
Carving	2026-10-26 04:01:00	recup_dir.1/ (E03)	138 files carved	Recovery of deleted remnants from unallocated space.

7. Conclusion and Limitations

RHINOUSB.dd holds deleted files in its unallocated space, including Word documents and images, even though only two files show up in the root directory. The three log files show web browsing sessions in April 2004. Those findings are solid.

Three limits apply to how far they can be pushed:

- **No attribution to a person.** Carved files and log entries show that data sat on a device or crossed a network. They do not show who was at the keyboard or who approved the activity.
- **Clocks can be wrong.** The August 2005 document dates and the April 2004 log dates rely on whatever clock each system was using, and either could have been changed or spoofed.
- **Not every artifact is deliberate.** Browser caching, background processes or a shared

machine could produce the same log entries and unallocated remnants without anyone choosing to store them.